Appendix A - Abrolhos to Eastern Recherche Natural Values Report

South-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table A1.1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
Dec 2014	2014-12 Geographe-bay stereo-BRUVs	BRUV	Coastal waters, Geographe HPZ, Geographe MUZ, Geographe NPZ, Geographe SPZ	324	320	297	238
Jun 2020	2020-06 south-west stereo-BRUVs	BRUV	South-west Corner NPZ	42	40	39	33
Oct-Nov 2020	2020-10 south-west stereo-BRUVs	BRUV	Coastal waters, South-west Corner NPZ, South-west Corner SPZ	275	273	272	244
Oct-Nov 2020	2020-10 south-west BOSS	BOSS	Coastal waters, South-west Corner NPZ, South-west Corner SPZ	262	262	-	-
Mar 2021	2021-03 Geographe BOSS	BOSS	Geographe HPZ, Geographe MUZ, Geographe NPZ, Geographe SPZ	198	198	-	-
Mar 2021	2021-03 West-Coast BOSS	BOSS	Coastal waters, South-west Corner NPZ	154	154	-	-
May 2021	2021-05 Abrolhos BOSS	BOSS	Abrolhos NPZ, Abrolhos SPZ	75	75	-	-
May 2021	2021-05 Abrolhos stereo-BRUVs	BRUV	Abrolhos NPZ, Abrolhos SPZ	50	50	49	47
Jan-Nov 2022	2022-11 Investigator stereo-BOSS	BOSS	Coastal waters, South-west Corner NPZ, South-west Corner SPZ	132	131	-	-
Nov 2022	2022-11 Investigator stereo-BRUVs	BRUV	South-west Corner NPZ	38	35	33	30
Nov-Dec 2022	2022-11 Salisbury stereo-BOSS	BOSS	Coastal waters, Eastern Recherche SPZ	118	118	-	-
Nov 2022	2022-11 Salisbury stereo-BRUVs	BRUV	Coastal waters, Eastern Recherche SPZ	40	40	40	38
Mar 2023	2023-03 Augusta stereo-BOSS	BOSS	Coastal waters, South-west Corner NPZ, South-west Corner SPZ	73	70	-	-

Mar 2023	2023-03 SwC BOSS	BOSS	South-west Corner NPZ	88	88	-	-
Mar 2023	2023-03 SwC stereo-BRUVs	BRUV	South-west Corner NPZ	32	32	32	32
May 2023	2023-05 Two-Rocks BOSS	BOSS	Coastal waters, Two Rocks MUZ, Two Rocks NPZ	137	137	-	-
Mar-Apr 2024	2024-04 Geographe stereo-BRUVs	BRUV	Coastal waters, Geographe HPZ, Geographe MUZ, Geographe NPZ, Geographe SPZ	261	260	259	246
Apr 2024	2024-04 Geographe BOSS	BOSS	Coastal waters, Geographe HPZ, Geographe MUZ, Geographe NPZ	121	121	-	-
May 2024	2024-05 Wudjari stereo-BRUVs	BRUV	Coastal waters, South-west Corner NPZ, South-west Corner SPZ	228	228	221	216
Oct 2024	2024-10 SWC stereo-BRUVs	BRUV	Coastal waters, South-west Corner NPZ, South-west Corner SPZ	190	190	188	178
Nov 2024	2024-11 SwC stereo-BOSS	BOSS	Coastal waters, Geographe MUZ, South-west Corner NPZ, South-west Corner SPZ	137	117	-	-
Feb-Mar 2025	2025-02 Yamatji-Shallow-Bank stereo-BRUVs	BRUV	Abrolhos NPZ, Abrolhos SPZ	120	115	114	109
Mar-Apr 2025	2025-04 South-west Corner BOSS	BOSS	South-west Corner NPZ, South-west Corner SPZ	210	210	-	-
Apr 2025	2025-04 South-west-corner stereo-BRUVs	BRUV	South-west Corner NPZ, South-west Corner SPZ	150	148	143	137
Jun-Jul 2025	2025-06 Wudjari stereo-BRUVs	BRUV	Eastern Recherche SPZ	120	120	118	111
Jun 2025	2025-06 Clio-Bank stereo-BRUVs	BRUV	Abrolhos SPZ	72	60	60	57

Benthic ecosystem extent

Observations

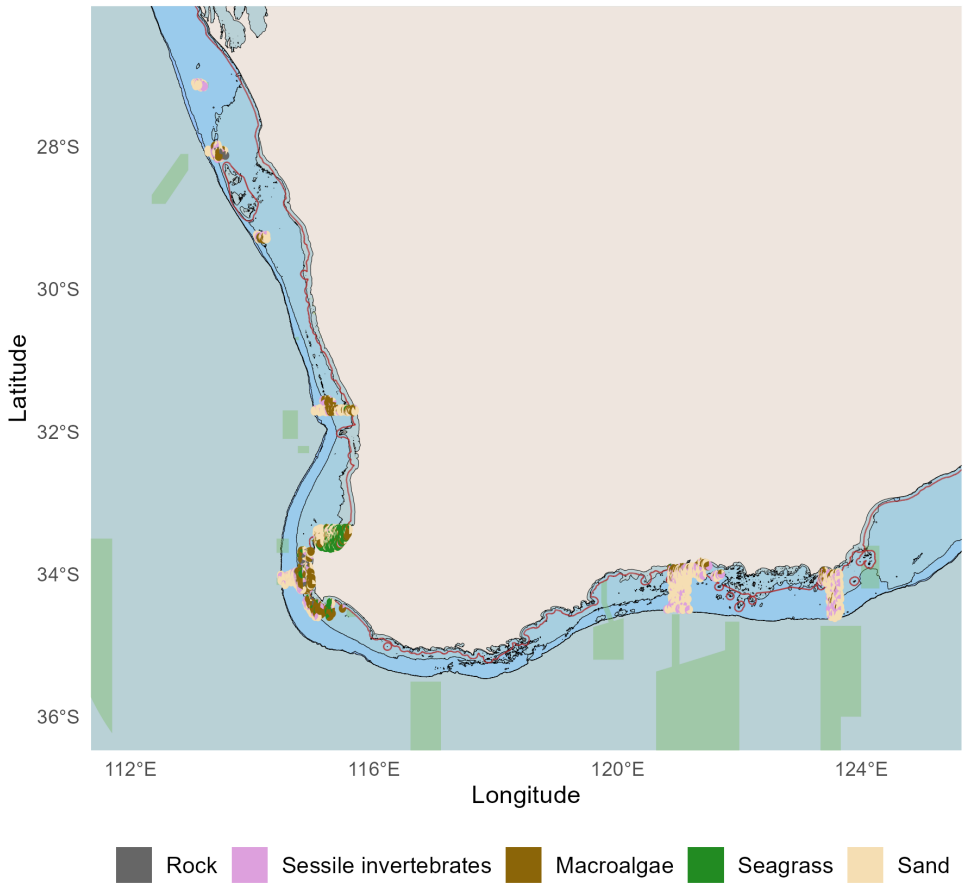

Figure A1.1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

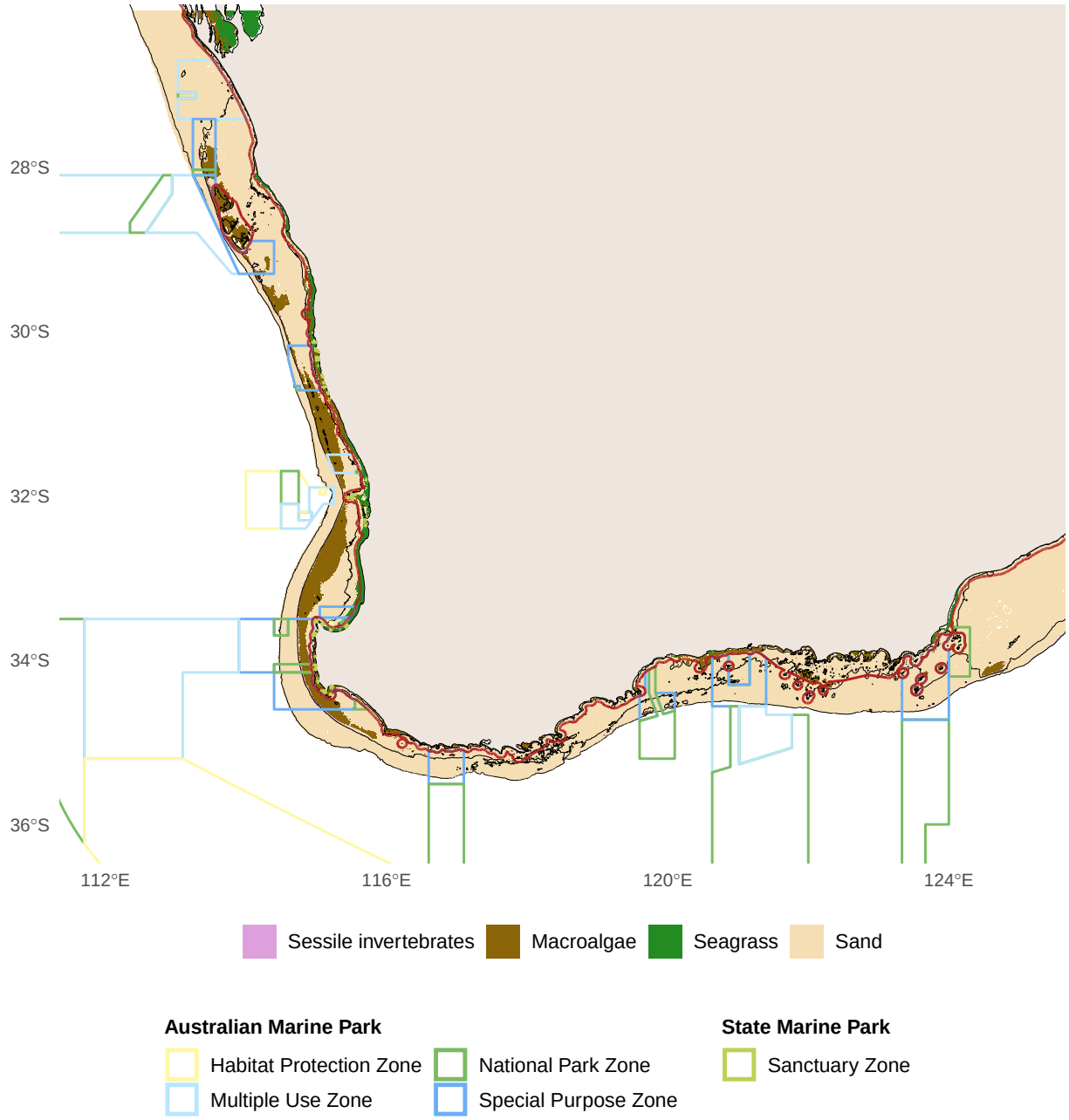

Figure A1.2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Funcional reef distribution and probability

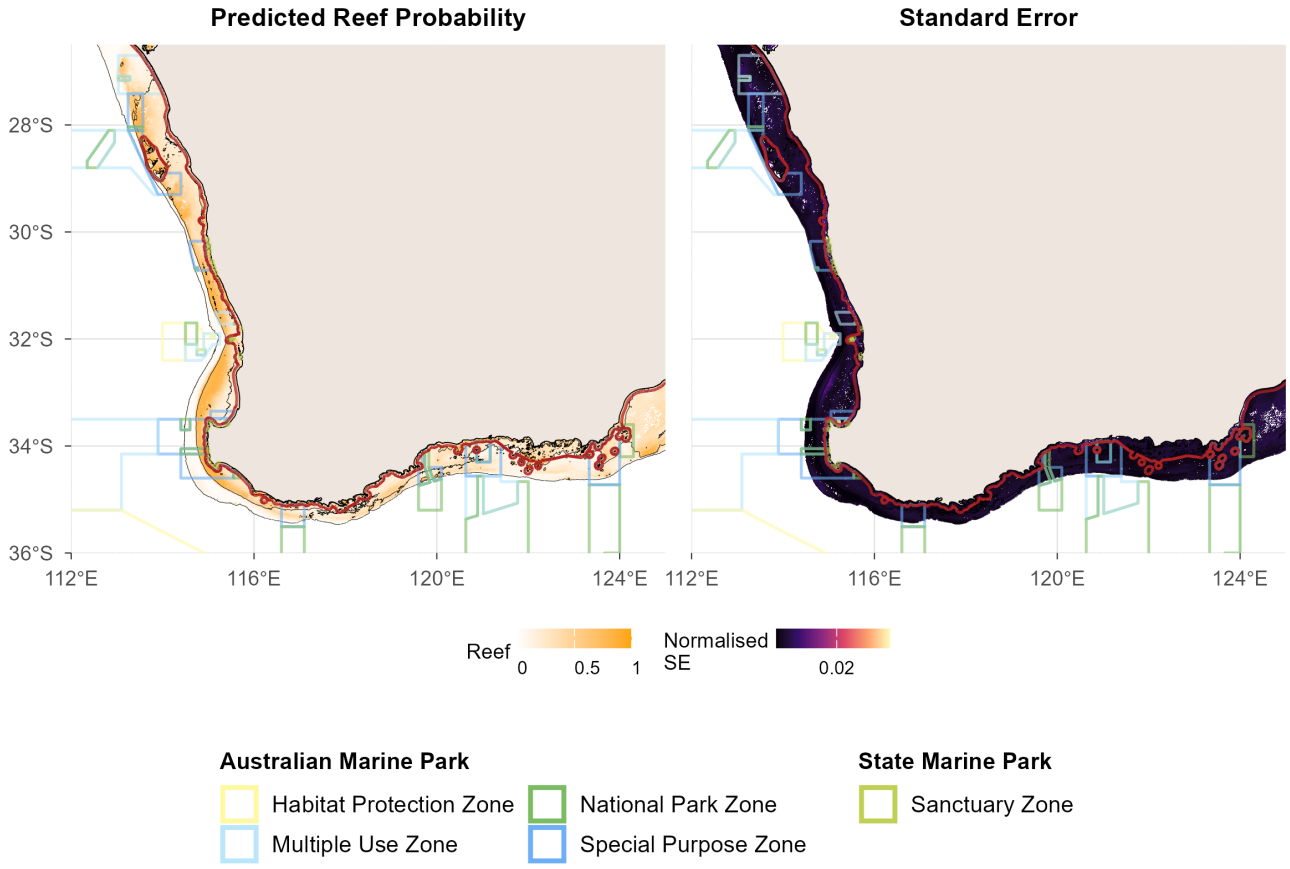

Figure A1.4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

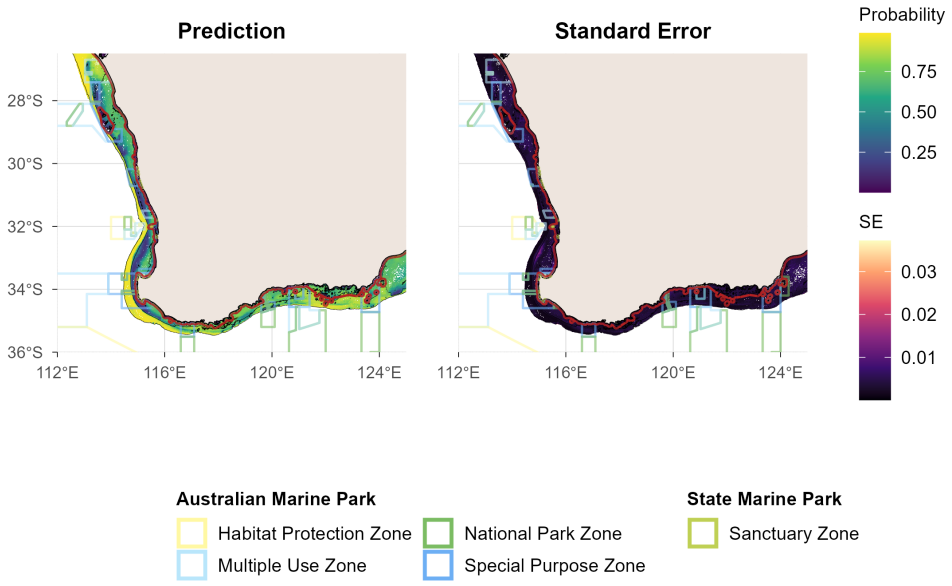

Figure A1.5.1: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

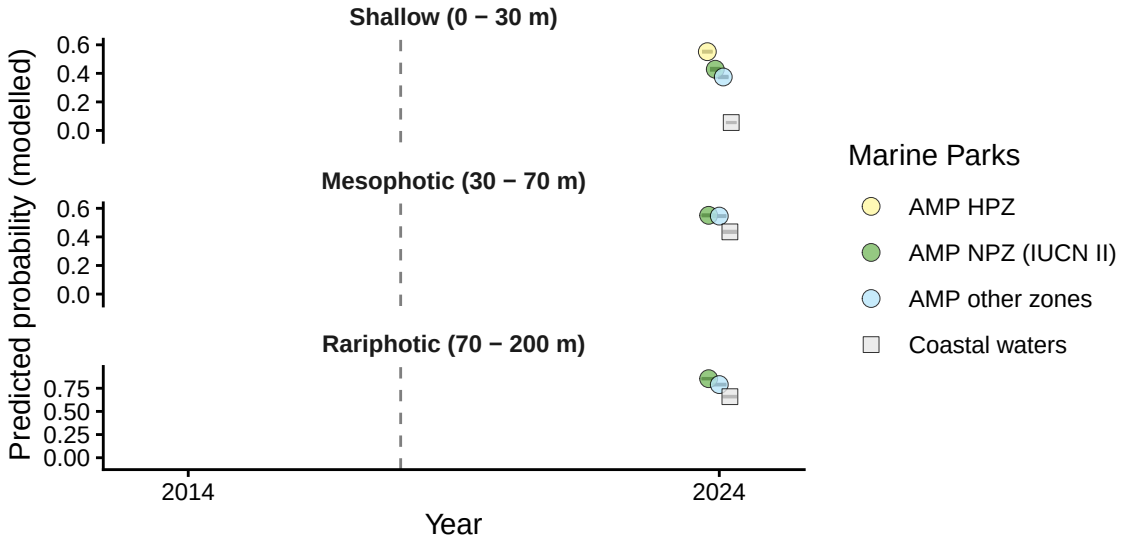

Figure A1.5.2: Sand: modelled (predicted) probability of occurrence from the final model, by zone and depth contour in 2024, the survey year with the most samples.

Macroalgae

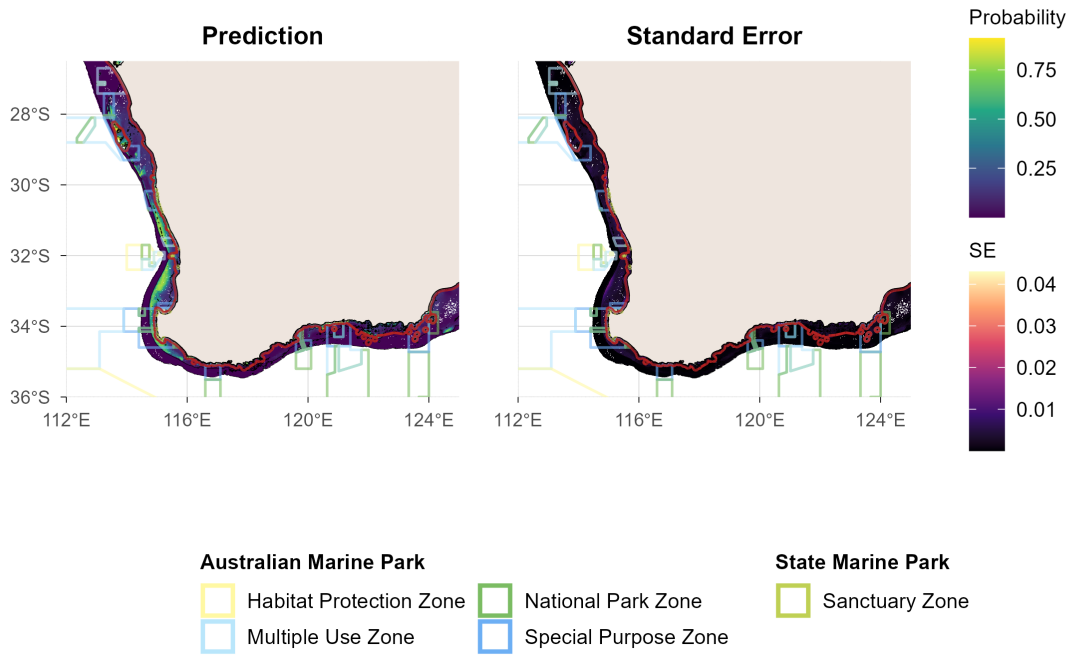

Figure A1.6.1: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

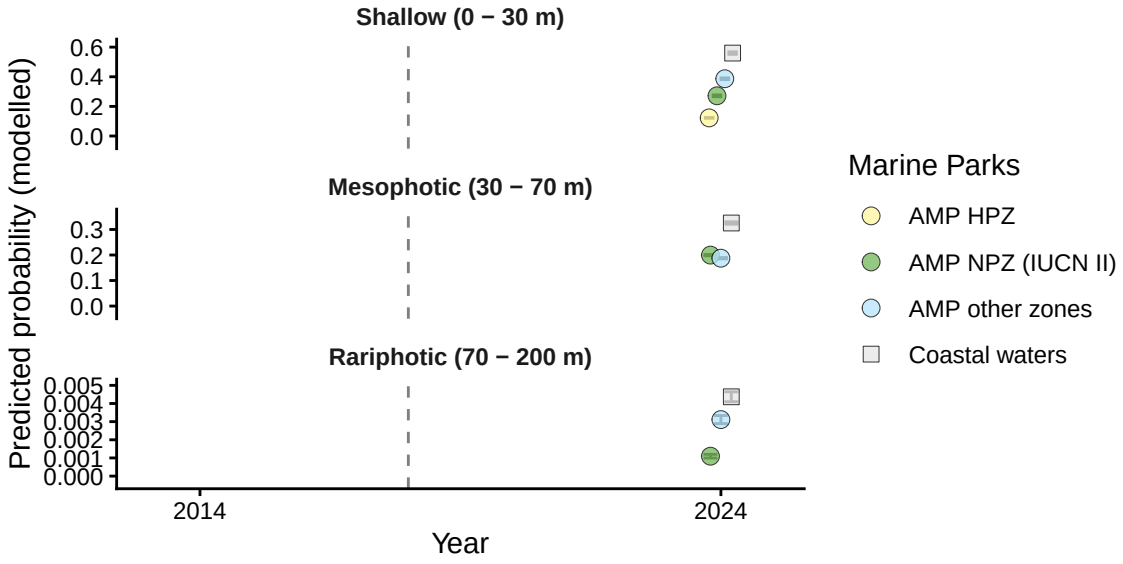

Figure A1.6.2: Macroalgae: modelled (predicted) probability of occurrence from the final model, by zone and depth contour in 2024, the survey year with the most samples.

Seagrass

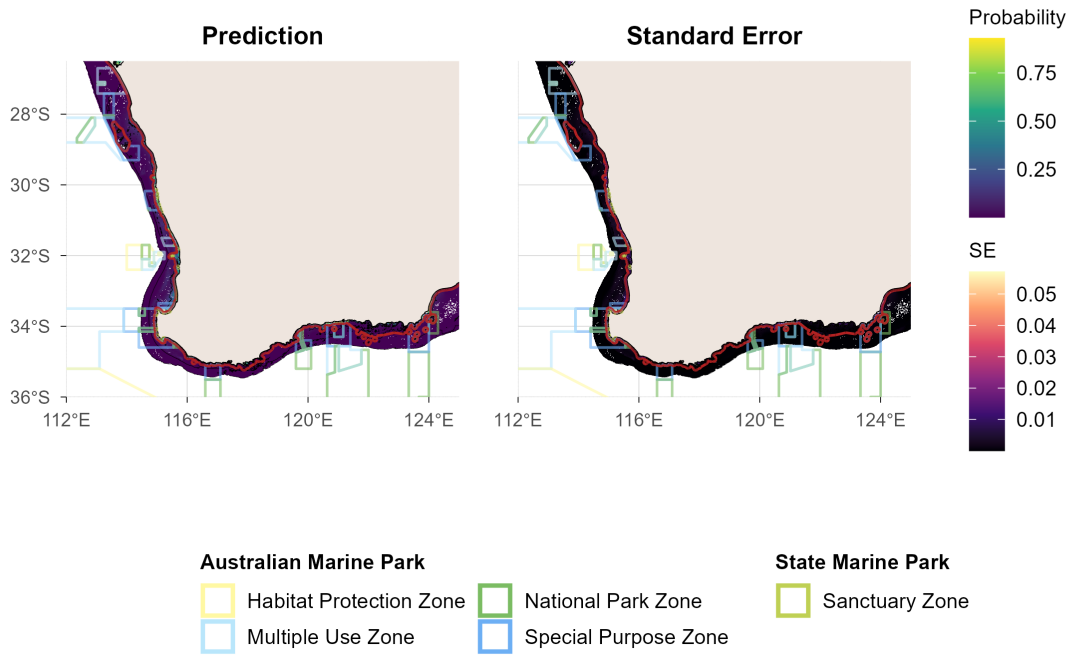

Figure A1.7.1: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

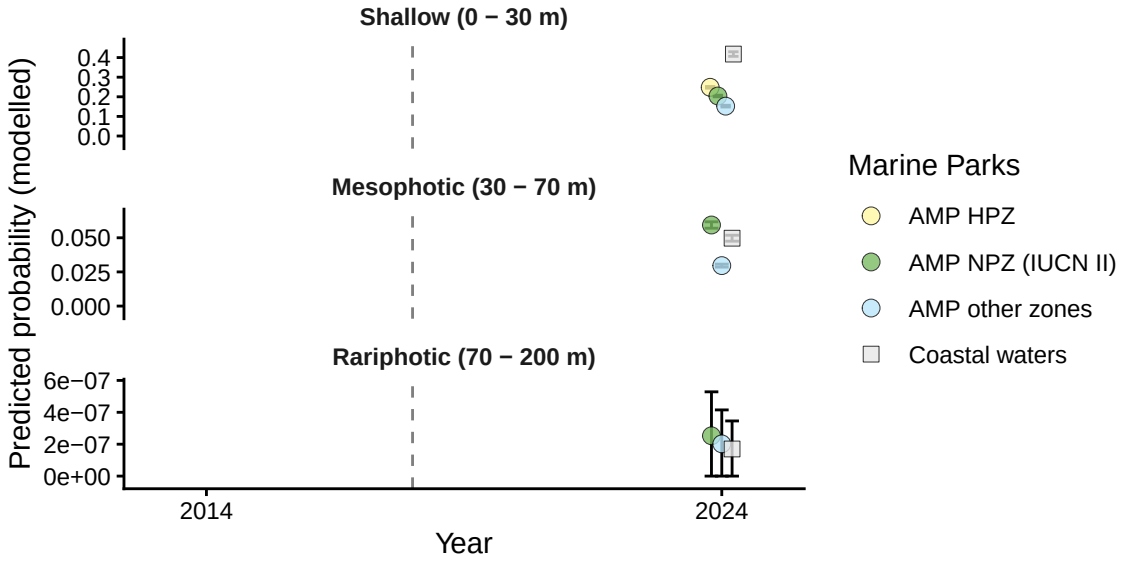

Figure A1.7.2: Seagrass: modelled (predicted) probability of occurrence from the final model, by zone and depth contour in 2024, the survey year with the most samples.

Sessile invertebrates

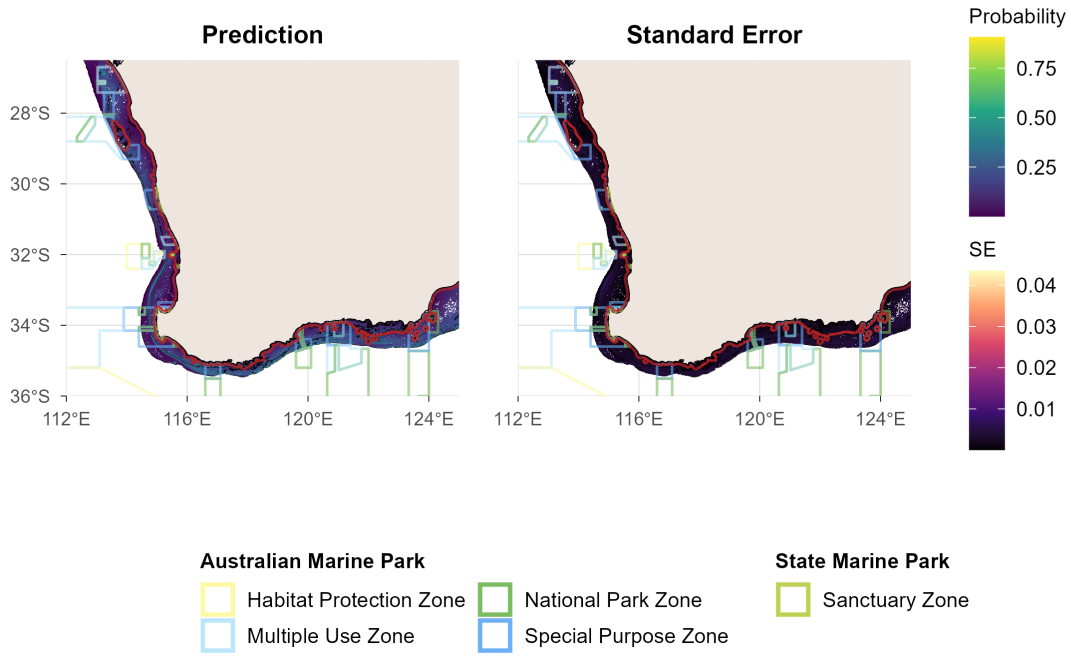

Figure A1.8.1: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

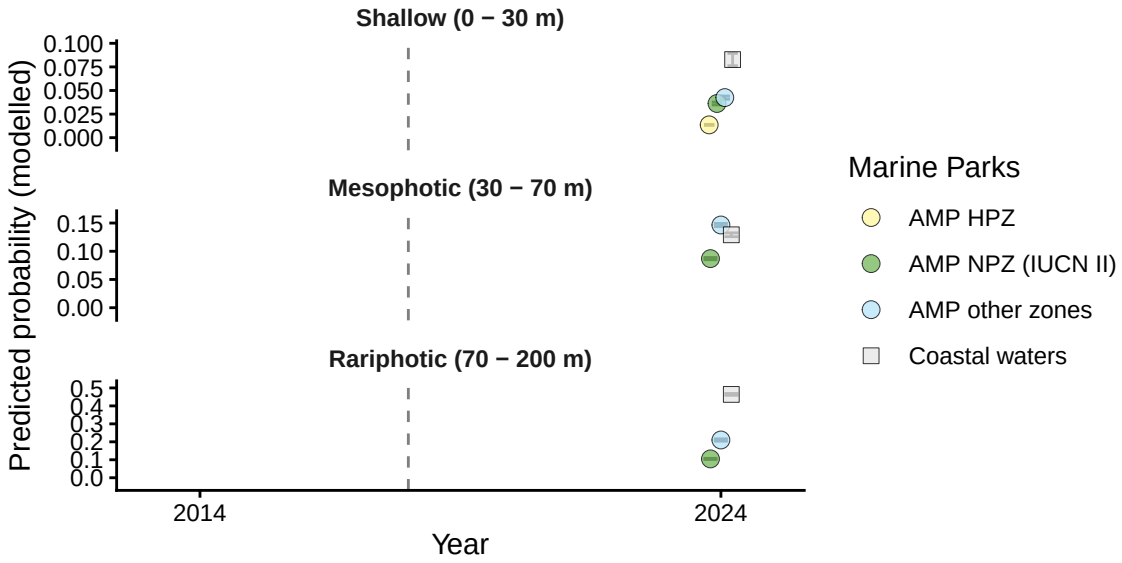

Figure A1.8.2: Sessile invertebrates: modelled (predicted) probability of occurrence from the final model, by zone and depth contour in 2024, the survey year with the most samples.

Rock

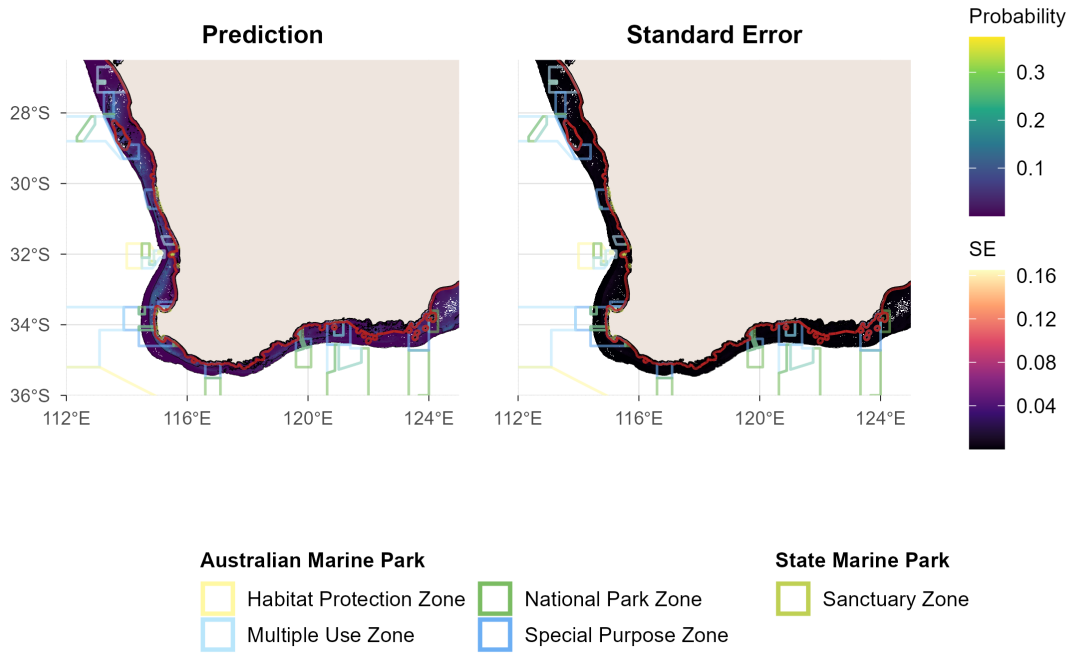

Figure A1.9.1: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

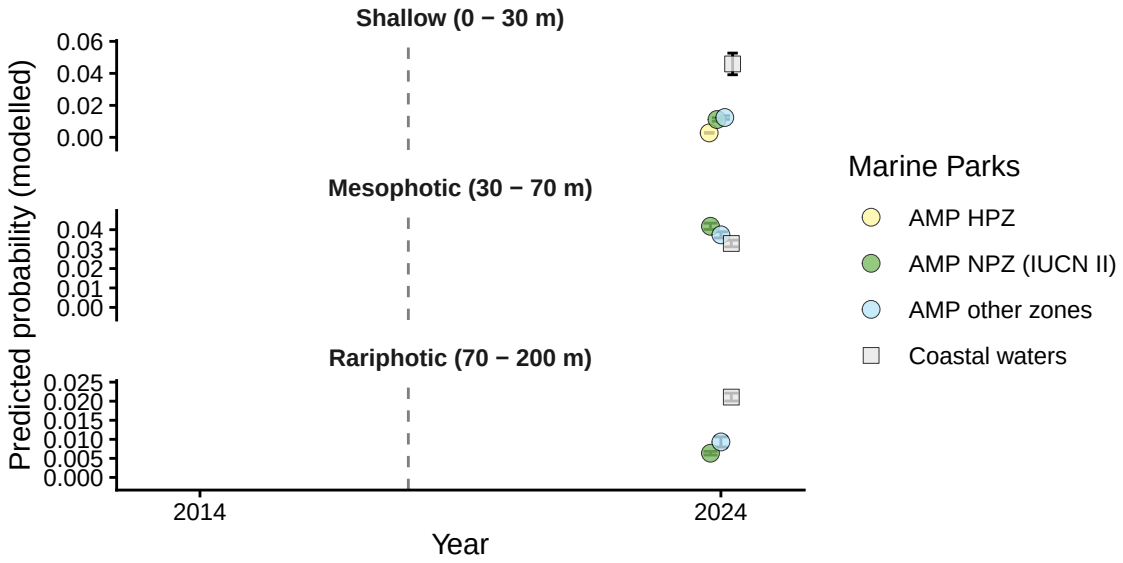

Figure A1.9.2: Rock: modelled (predicted) probability of occurrence from the final model, by zone and depth contour in 2024, the survey year with the most samples.

Benchmarks of demersal fish natural values

Survey effort

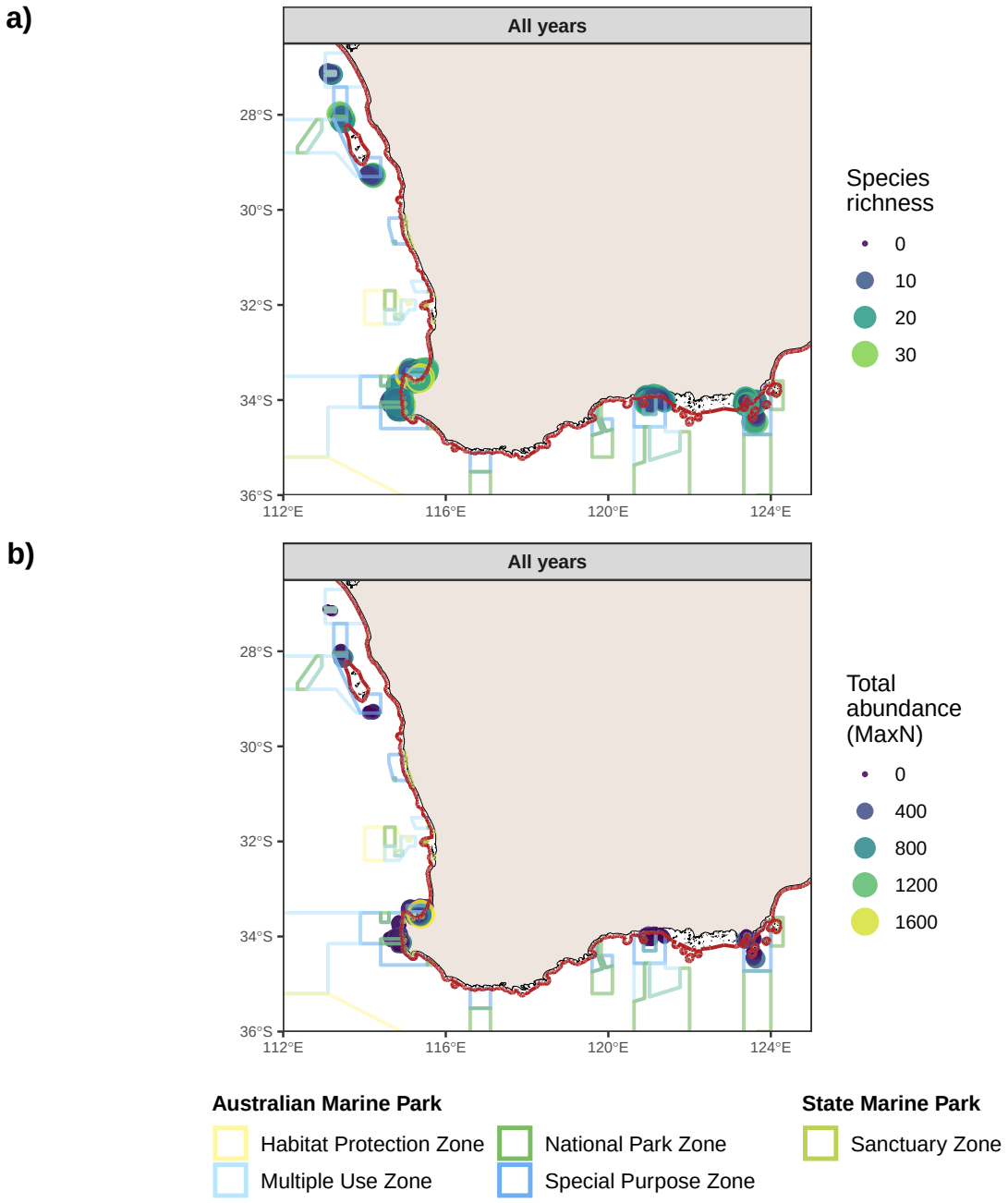

Figure A1.10: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

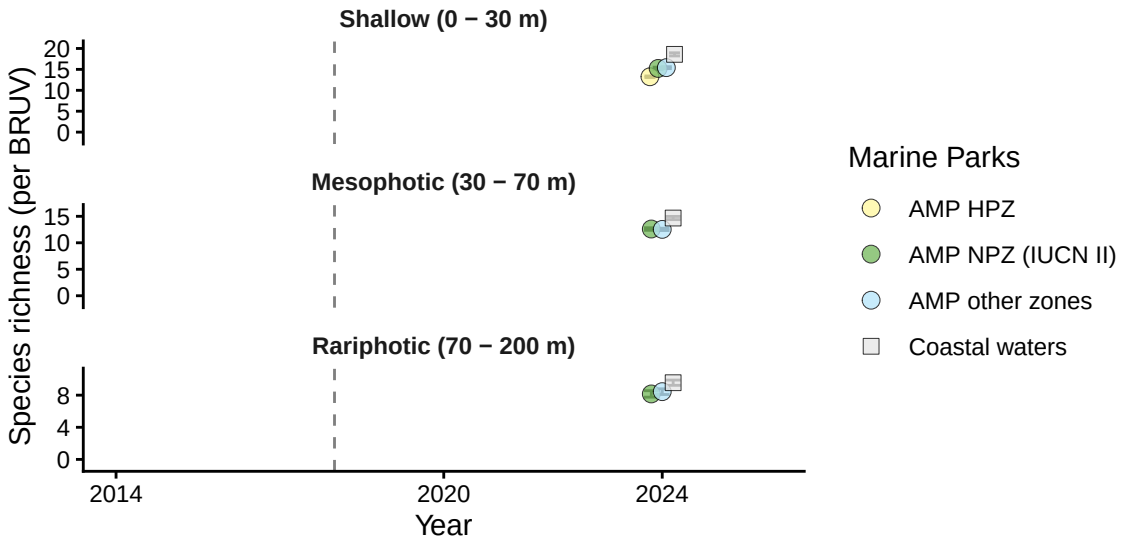

Figure A1.11.1: Demersal fish - Species richness: modelled (predicted) values from the final model, by zone and depth contour in 2024, the survey year with the most stereo-BRUV deployments.

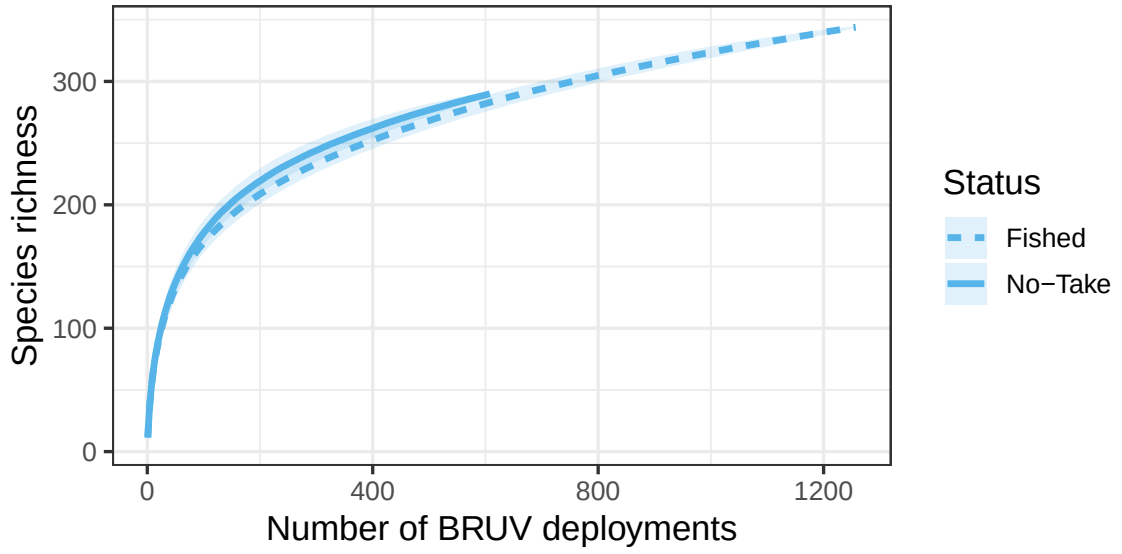

Figure A1.11.2: Demersal fish - Species richness: species accumulation curve (all years pooled) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Species richness

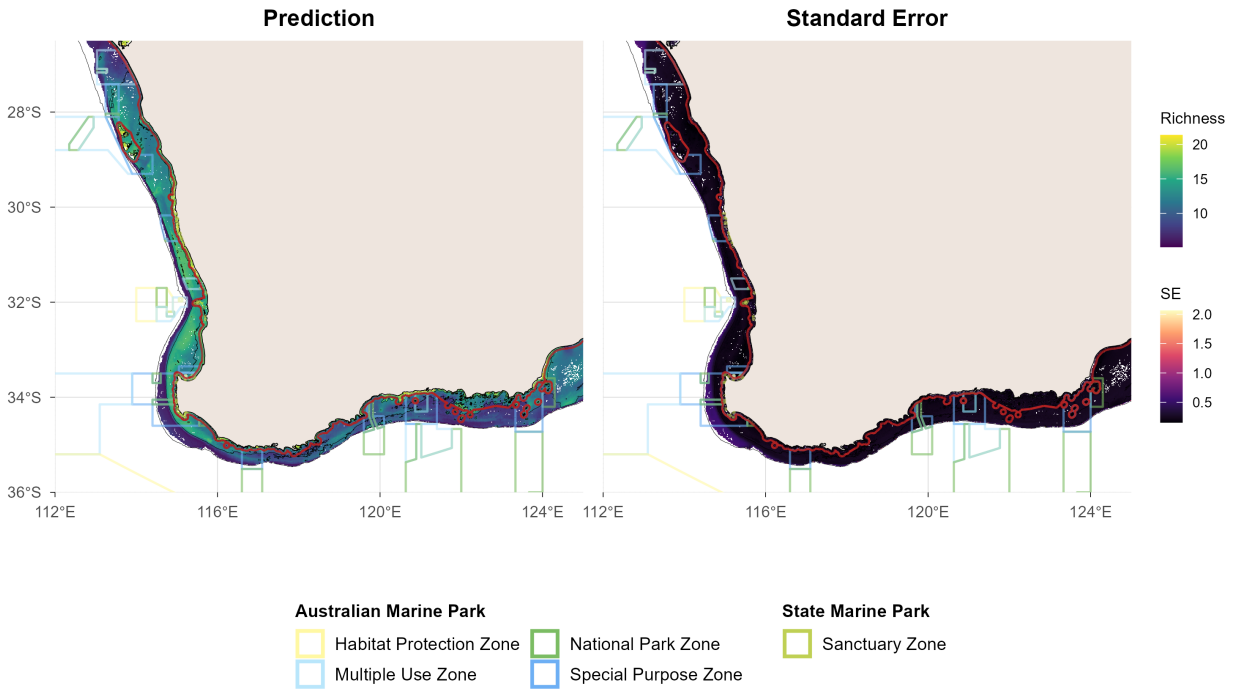

Figure A1.11.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

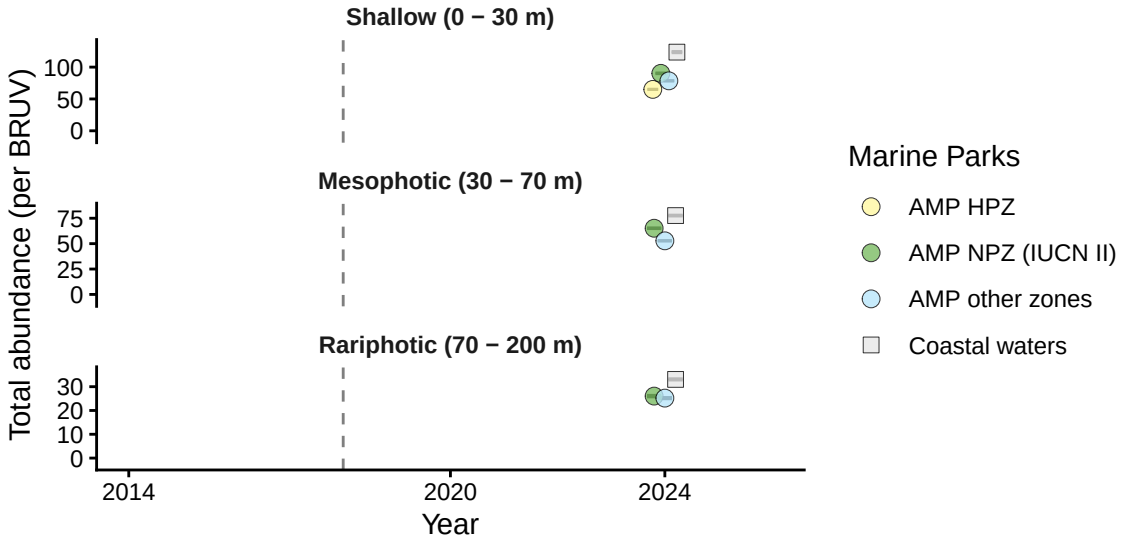

Figure A1.12.1: Demersal fish - Total abundance: modelled (predicted) values from the final model, by zone and depth contour in 2024, the survey year with the most stereo-BRUV deployments.

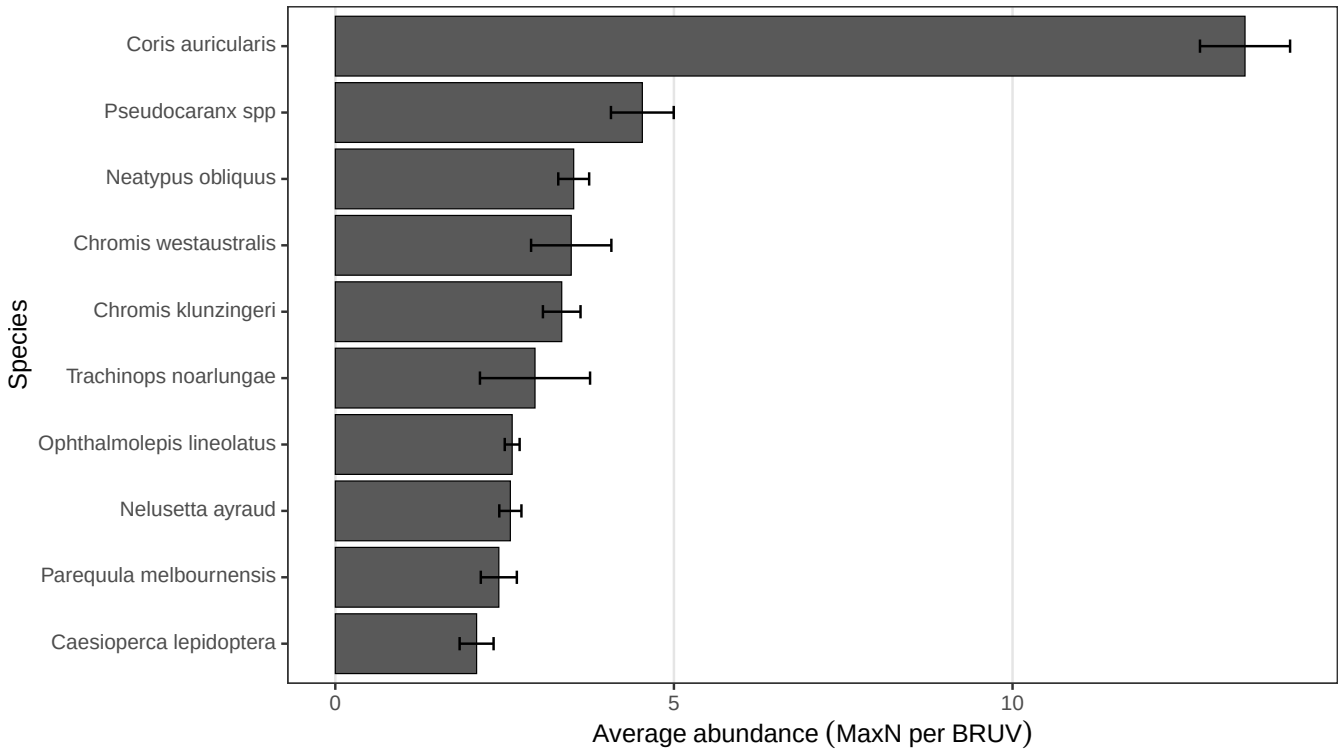

Figure A1.12.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Total abundance

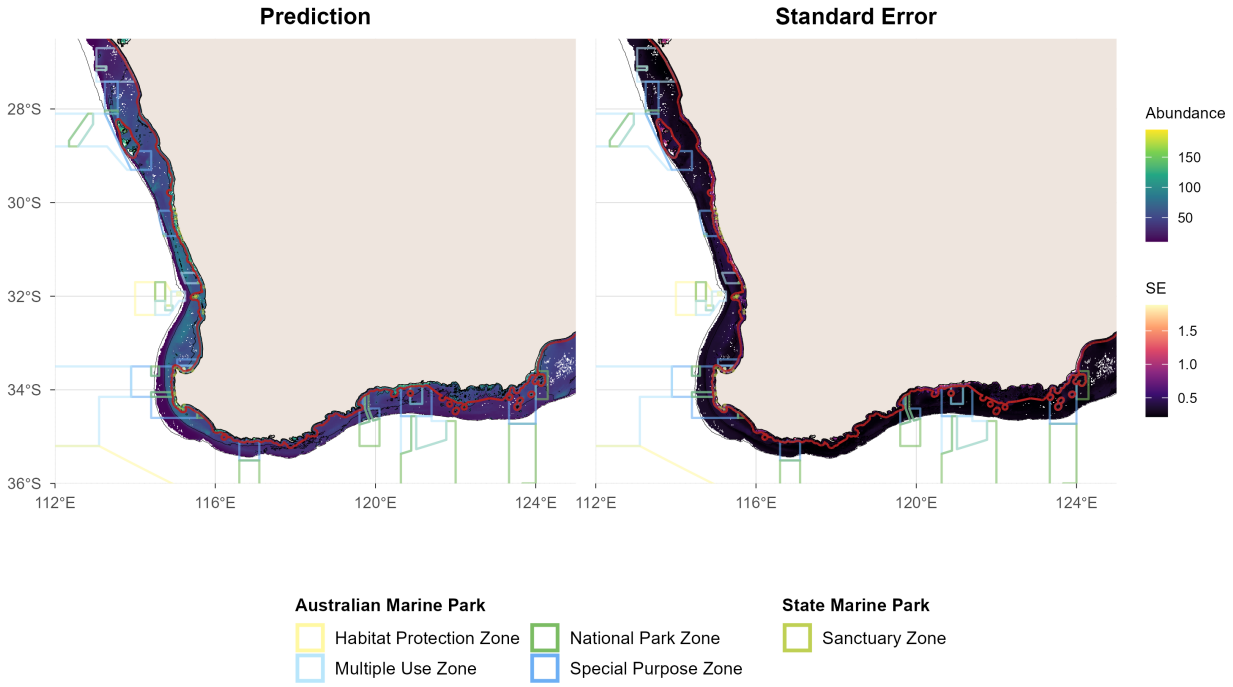

Figure A1.12.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

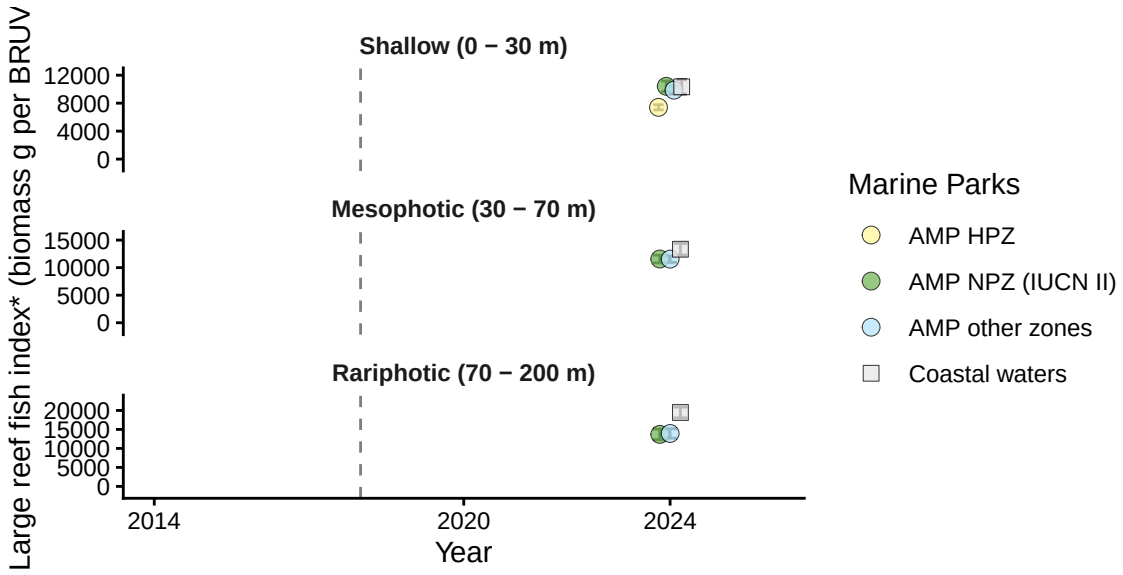

Figure A1.13.1: Demersal fish - Large Reef Fish Index*: modelled (predicted) values from the final model, by zone and depth contour in 2024, the survey year with the most stereo-BRUV deployments. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

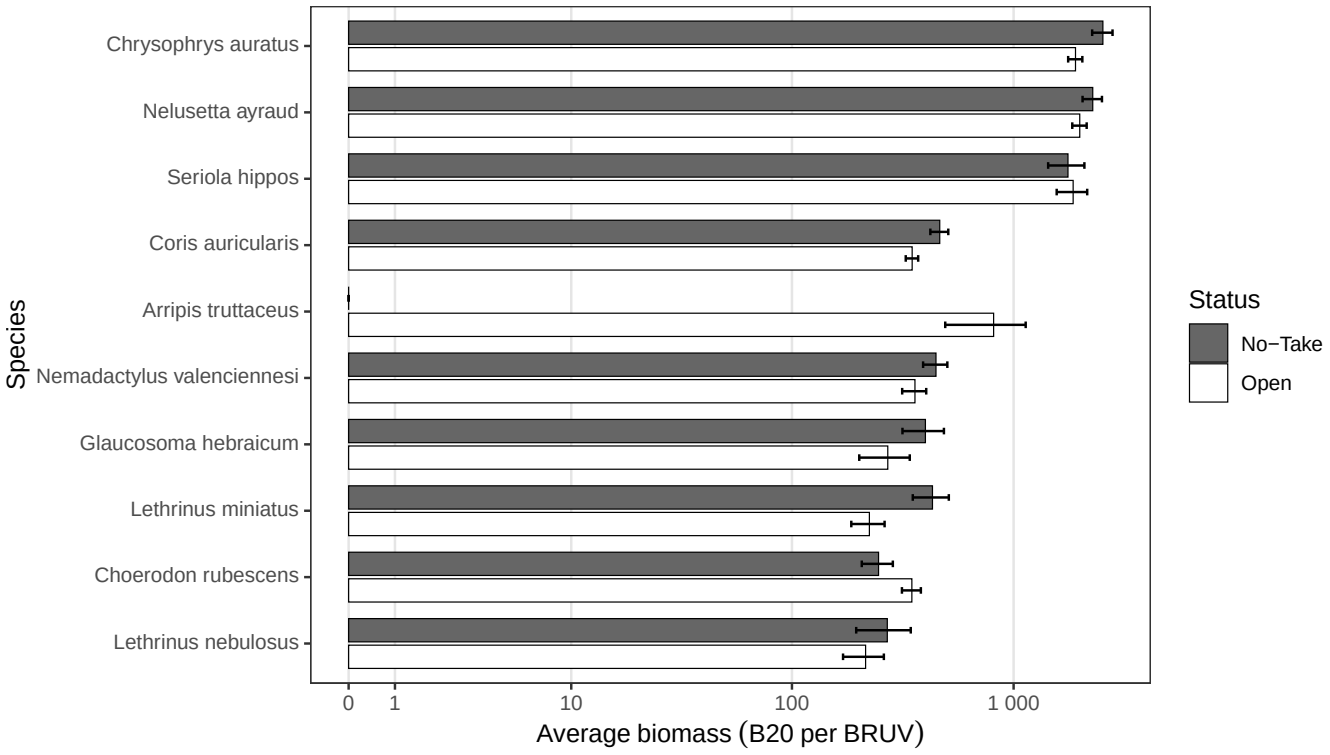

Figure A1.13.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Large Reef Fish Index*

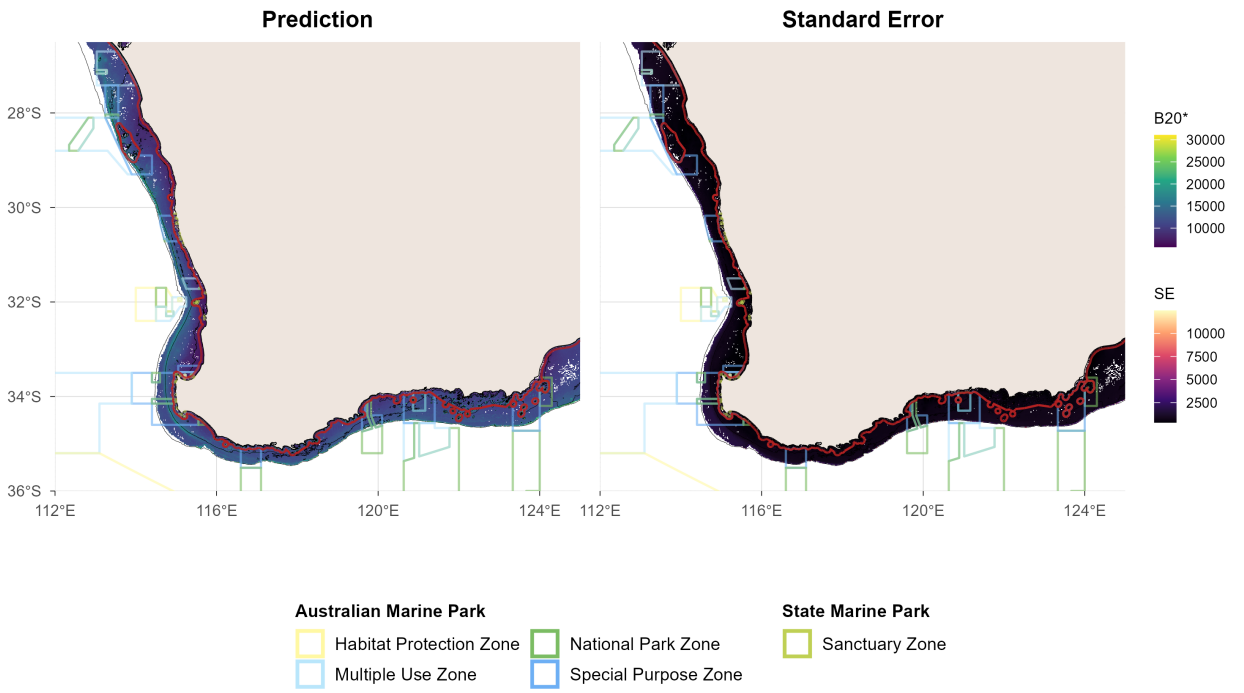

Figure A1.13.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

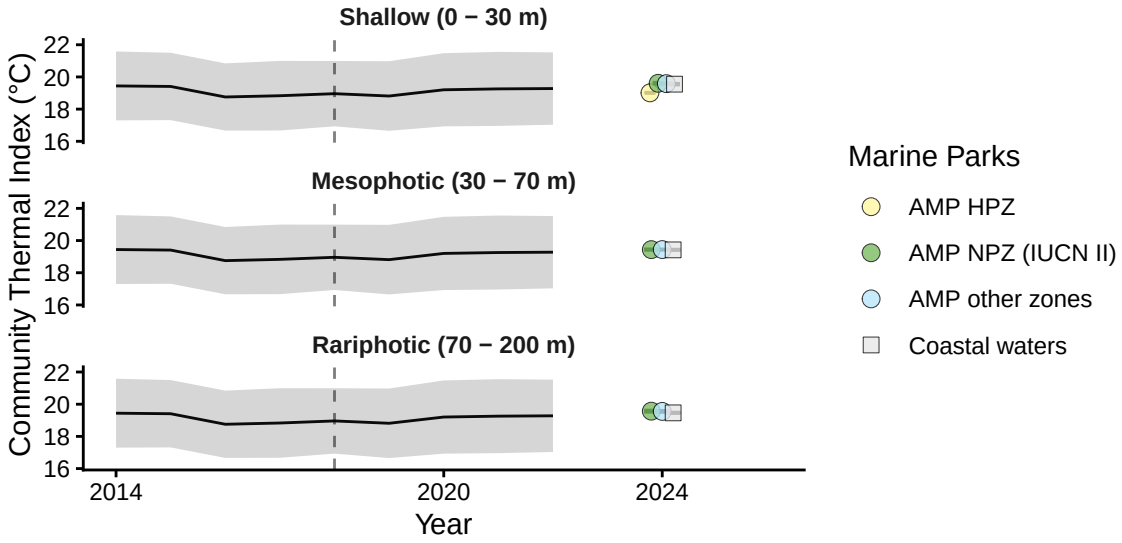

Figure A1.14.1: Demersal fish - Reef Fish Thermal Index: modelled (predicted) values from the final model, by zone and depth contour in 2024, the survey year with the most stereo-BRUV deployments, with sea surface temperature (black line, grey band = SD) overlaid.

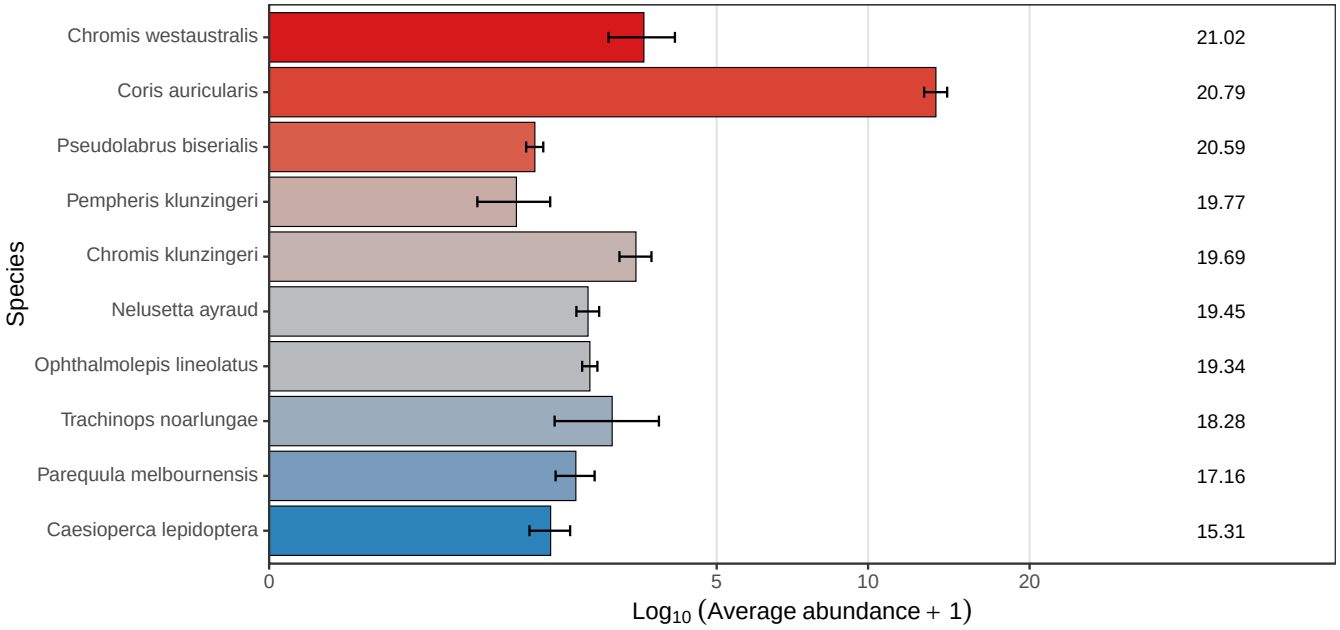

Figure A1.14.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Reef fish thermal index

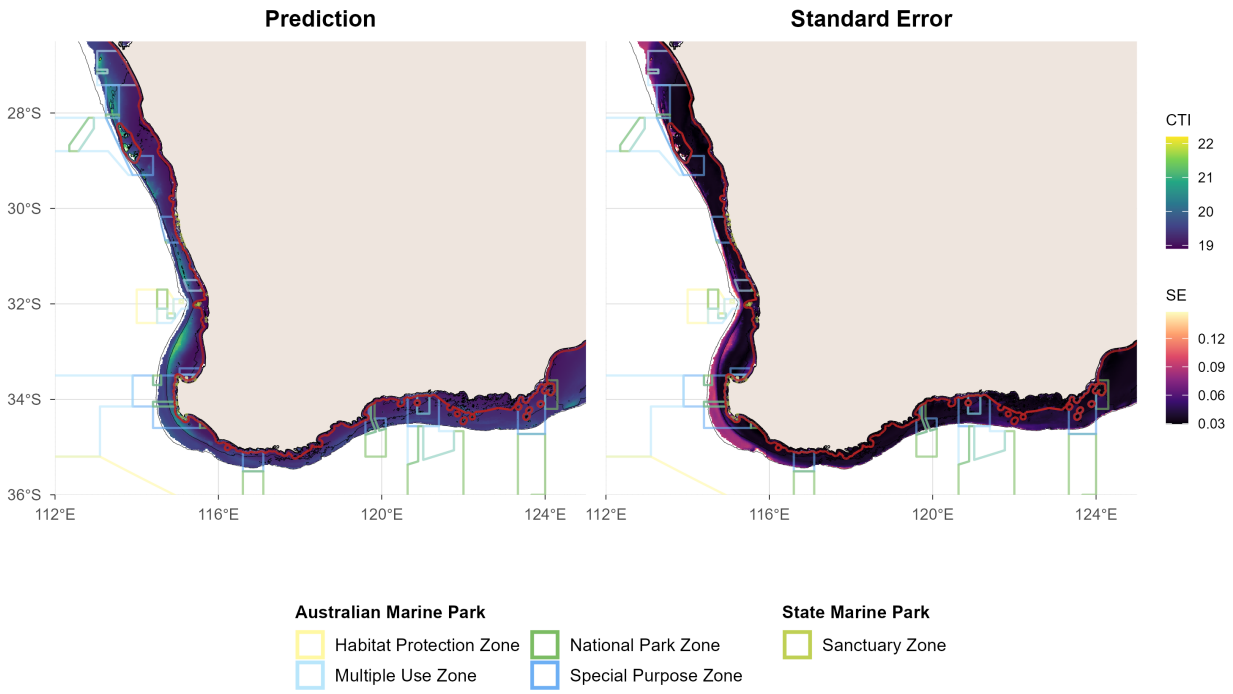

Figure A1.14.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Threatened species

Table A1.2: Threatened species recorded in stereo-BRUV imagery across all survey years, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. *This species is also listed as 'Conservation dependent' on the EPBC Act List of Threatened Fauna. ^This species is also listed as 'Vulnerable' on the EPBC Act List of Threatened Fauna.

Species	Common name	Family	IUCN category	Samples	Total MaxN	Years recorded
Epinephelides armatus	Breaksea Cod	Serranidae	Near Threatened	222	319	2014, 2020, 2022, 2023, 2024, 2025
Bodianus frenchii	Foxfish	Labridae	Near Threatened	207	267	2014, 2020, 2022, 2023, 2024, 2025
Sepia apama	Giant cuttlefish	Sepiidae	Near Threatened	61	70	2024, 2025
Achoerodus gouldii	Western Blue Groper	Labridae	Vulnerable	47	51	2014, 2020, 2022, 2023, 2024, 2025
Carcharhinus plumbeus	Sandbar Shark	Carcharhinidae	Endangered	34	35	2020, 2021, 2024, 2025
Carcharhinus brevipinna	Spinner Shark	Carcharhinidae	Vulnerable	24	36	2014, 2020, 2023, 2024, 2025
Carcharhinus obscurus	Dusky Whaler	Carcharhinidae	Endangered	20	23	2022, 2024, 2025
Squalus acanthias	Whitespotted Dogfish	Squalidae	Vulnerable	18	28	2020
Galeocerdo cuvier	Tiger Shark	Carcharhinidae	Near Threatened	15	15	2014, 2021, 2024, 2025
*Sphyrna zygaena**	Smooth Hammerhead	Sphyrnidae	Vulnerable	13	13	2014, 2020, 2022, 2024
Rhynchobatus australiae	Whitespotted Guitarfish	Rhinidae	Critically Endangered	6	6	2021, 2024, 2025
*Sphyrna lewini**	Scalloped Hammerhead	Sphyrnidae	Critically Endangered	6	175	2021, 2024, 2025
Carcharhinus limbatus	Common Blacktip Shark	Carcharhinidae	Vulnerable	5	7	2020, 2024
Loxodon macrorhinus	Sliteye Shark	Carcharhinidae	Near Threatened	5	6	2025
Taeniurops meyeni	Blotched Fantail Ray	Dasyatidae	Vulnerable	4	4	2021, 2025
*Galeorhinus galeus**	School Shark	Triakidae	Critically Endangered	3	3	2020, 2024
Carcharhinus albimarginatus	Silvertip Shark	Carcharhinidae	Vulnerable	2	2	2025
Carcharodon carcharias^	White Shark	Lamnidae	Vulnerable	2	2	2024, 2025
Isurus oxyrinchus	Shortfin Mako	Lamnidae	Endangered	2	2	2014, 2025
*Thunnus maccoyii**	Southern Bluefin Tuna	Scombridae	Endangered	2	2	2024
Bathytoshia lata	Black Stingray	Dasyatidae	Vulnerable	1	1	2024
Carcharhinus altimus	Bignose Shark	Carcharhinidae	Near Threatened	1	2	2025
Carcharhinus brachyurus	Bronze Whaler	Carcharhinidae	Vulnerable	1	1	2024
Carcharhinus sorrah	Spot-tail Shark	Carcharhinidae	Near Threatened	1	1	2025
Carcharias taurus	Greynurse Shark	Carchariidae	Critically Endangered	1	1	2024
Neophoca cinerea	Australian sea-lion	Otariidae	Endangered	1	1	2025

Threatened shark and ray species

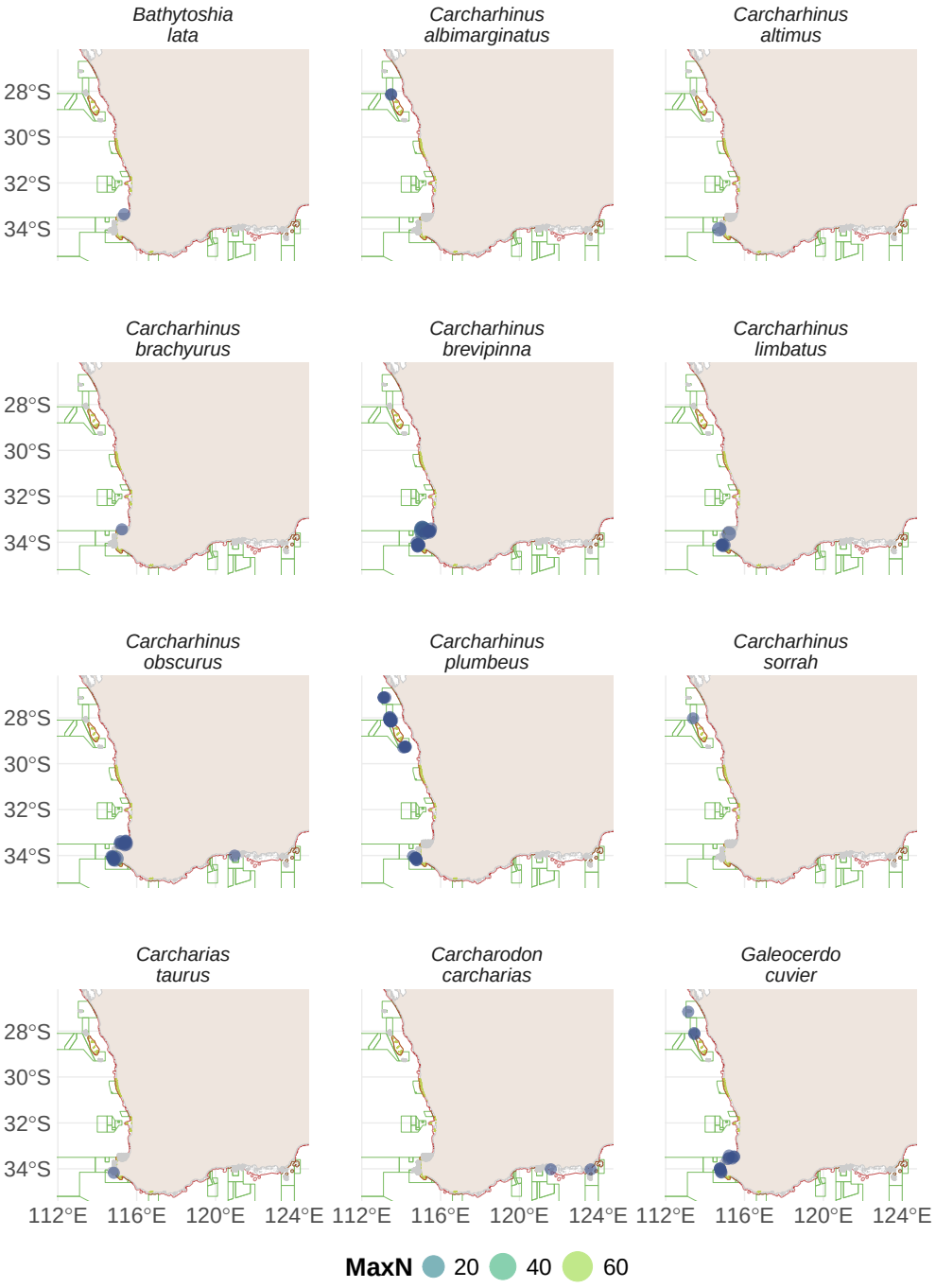

Figure A1.15.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery. Points are scaled and coloured by abundance (MaxN); grey points show every stereo-BRUV deployment. Commonwealth and coastal marine park boundaries are shown. The red line delimits coastal and Commonwealth waters.

Other shark and ray species

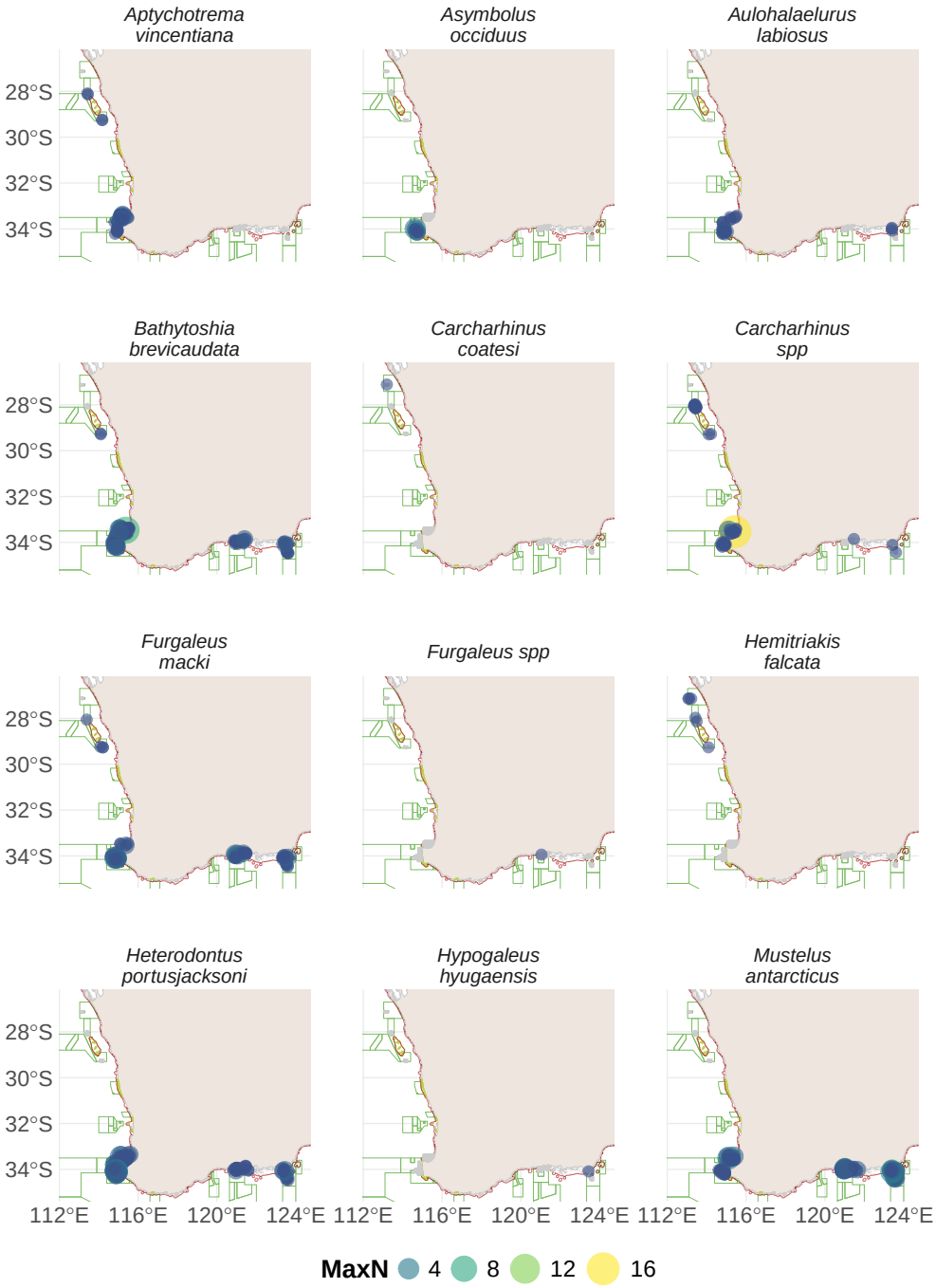

Figure A1.15.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery. Points are scaled and coloured by abundance (MaxN); grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth and state marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species

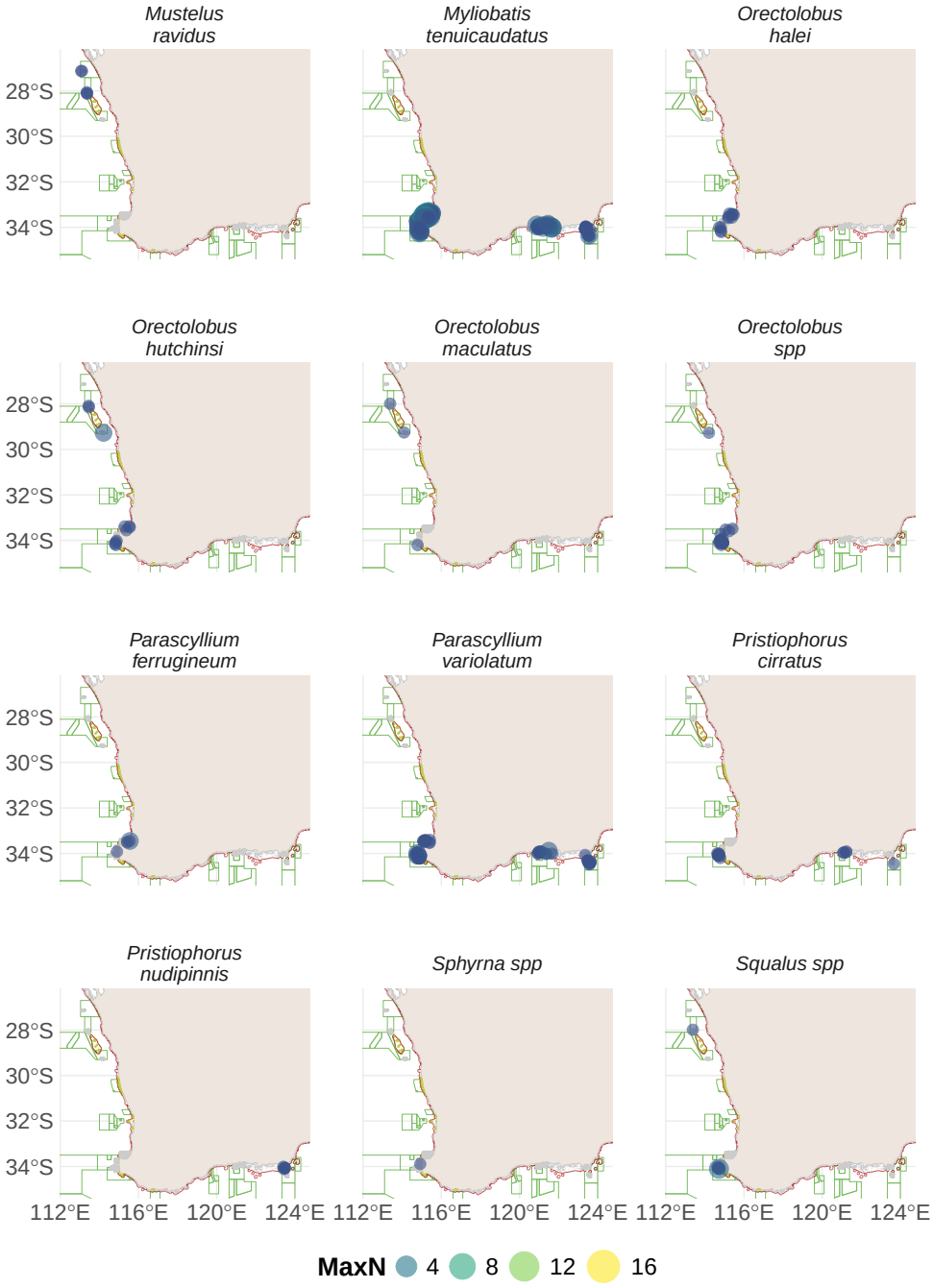

Figure A1.15.3: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery (continued). Points are scaled and coloured by abundance (MaxN); grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth and state marine park boundaries are shown. The red line delimits state and commonwealth waters.